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Powder Characterization & Effect of reuse powders on part properties in Laser Additive Manufacturing (LAM)

Submitted in partial fulfilments of requirements

of the degree of

Master of Technology

in

Additive Manufacturing

by

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**DEPARTMENT OF MECHANICAL ENGINEERING
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June, 2018

APPROVAL SHEET

This dissertation entitled “**Powder Characterization & effect of reuse powders on part properties in Laser Additive Manufacturing**” by **Mr. Kritagya Kumar, Roll No. 163804** is approved for the degree of Master of Technology in Additive Manufacturing.

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DECLARATION

I declare that this written thesis submission represents my work in my own words and where other's ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or exaggerate any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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CERTIFICATE

This is to certify that the dissertation work entitled **“Powder Characterization & effect of reuse powders on part properties in Laser Additive Manufacturing”** is a bonafide record of work carried out by **Mr. Kritagya Kumar, Roll No. 163804**, submitted to the faculty of **“Mechanical Engineering Department”**, in partial fulfilment of the requirements for the award of the degree of **“Master of Technology in Additive Manufacturing”** at National Institute of Technology, Warangal during the academic year 2016–2018.

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About the organisation

The wipro3D is a unit of Wipro Enterprises Limited, engaged in design and manufacturing house, which offers end-to-end Design For Additive Manufacturing (DFAM), Prototyping, Build and Production Services for demanding applications in Aerospace, Space, Defence, Oil & Gas, Industrial, Tooling, Automotive and Medical segments. The organization is focused on leading edge work for industry leading clients to provide Design & Additive Manufacturing solutions for critical parts.



Figure 1 Wipro3D

Research activities at Wipro3D

- Parameter Development for new materials
- Shrinkage Control
- Impact of recoater speed on material properties
- Residual Stress management in platform
- Platform selection based on exposure area
- Effect of gas flow

Aim of Wipro3D

WIPRO3D delivers Additively Manufactured components in production level volumes, using a complete “Select-to-Prove” lifecycle approach, based on extensive experience in Component Selection and use case analysis, Additive Engineering Design and simulation, Material Adaption and Substitution, Build Strategies and Post Processing know-how. Value delivery remains a core consideration in our solutions and services, supported by a strong quality framework.

Mission of Wipro3D

Laser Additive Manufacturing Lab. totally focuses on the area of additive manufacturing to conduct cutting edge research with the use of *Design Methods, Manufacturing and Materials Processing Technologies and Materials Science and Testing*.

Facilities available in Wipro3D

All research work in Wipro3D including *Designing, Manufacturing component, Material Science, Powder Mixing, Process Optimization, Analysis, Testing* are accomplished using the following in-house developed facilities:

1. EOS M290, M280, M400
2. Automatic powder handling module
3. Inert Gas & Vacuum furnace
4. 4 axis Vertical Machining Centre
5. Shot peening (various media)
6. EDM Wire Cut

7. Surface Grinder

- 8. Specialized Machining processes**
- 9. Tumbling Equipment**
- 10. Particle size analyser**
- 11. Optical Microscope**
- 12. 7 Axis 3D Scanner**
- 13. ONH Analyser**
- 14. CMM**
- 15. Part Density Measurement System**
- 16. Powder Flow ability Test Equipment**

ABSTRACT

Additive manufacturing is a suitable process to have high strength components with complex geometry. In order to produce structures by laser additive manufacturing of sufficient standard and also meet the desired properties, powder quality should be properly checked. It is therefore of importance to understand the impact of various powder characteristics on the final manufactured part. The knowledge of powder characteristics and their correlation to the sintered layer is important development of the technology in order to enhance future production capabilities. Nickel based superalloys and titanium are commonly used materials for high-performance applications and these materials are considered in this project.

Metal powder re-use in additive manufacturing (AM) has been investigated using an EOS M290 system for titanium alloy Ti-6Al-4V powder and Inconel 718 was studied on EOS M400. The study is to characterize the powder and effect of various powder characteristics on their part property in which the chemical and physical effects of re-using metal powder in these machines. In order to increase the robustness of the study, hardness coupons were built for each run and same coupons were used for density measurement as well.

A single batch of powder was cycled through EOS's additive manufacturing process for a total of 6 builds for Inconel 718 and 4 builds for Ti64. Powder was analyzed for oxygen and nitrogen content, Hall flow, particle size distribution (PSD) and density. There was found to be a small but steady increase of both oxygen and nitrogen concentrations in the powder as the build number increased. Oxygen level content after 3rd build start to fluctuate but it was in the limit given in the material specification sheet. It has been also observed that regular topping of fresh powder to the dispenser is helpful in controlling the limit. PSD values were found to be changed as smaller particles has decreased. Hardness and density for the part was constant. Overall, it was concluded that the changes in the powder over the period of the study are not significant enough to affect the material parameter settings, and there is no evidence to suggest for powder disposal.

Keywords: Powder Characteristics, Laser Additive Manufacturing (LAM), Inconel 718, Ti64, Particle Size Distribution (PSD), ONH, Hall Flow meter, Carney Flow meter

AIM AND SCOPE

The main objective of this thesis work is to analyze powder characterization after every build and changes in powder during these builds. To achieve this objective, ASTM Standard for powder characterization were followed in which ONH, Dry sieving, Flow ability, particle size distribution was checked using their respective standard. For part quality true density, microstructure and hardness were also been checked. The aim of this project is to find a powder characteristics guideline, based on analysis done, to review on powder batches to judge an output of certain quality. The Main objective to find whether some powder characteristics have an effect on part quality?

The scope of this work is to introduce distinguish and evaluate powder characteristics of interest. For this by laser metal sintering of standardized structures was used and their final quality was evaluated. The aim is to identify and discuss any correlation between the characteristics of powder and final part quality.

ORGANISATION OF THESIS

This thesis is divided into five chapters. The topic and aim of the research work will be introduced in chapter one along with significant contribution and thesis organization. A comprehensive review on literature is presented in chapter two. The third chapter outlines the material studies and experimental methods related to this thesis. The fourth chapter outlines the experimental work. The results of the experiments are reported in the fifth chapter. Finally, the concluding remarks are provided in the final chapter six in the form of conclusion and recommendation for future work.

Abbreviations & Nomenclature

Name	Abbreviation
Laser Additive manufacturing	LAM
Inconel 718	IN718
Ti6Al4V	Ti64
American society for testing and materials	ASTM
Electrode induction melting gas atomization	EIGA
Face-centered cubic	FCC
Gas atomization	GA
Gas atomization	GA
Particle size distribution	PSD
Plasma Atomization	PA
Plasma rotation electrode process	PREP
Nickel	Ni
Oxygen Nitrogen Hydrogen	ONH

Parameter	Symbol	Unit
Apparent density		g/cm^3
Part density		g/cm^3
Flow rate index	FRI	-
Hall Flow rate		s/50g
Heat flux		W/m^2
Laser power		W
Powder efficiency		%
Powder mass delivered		g
Powder mass deposited		g
Powder mass flow rate		g/min
Scanning speed		mm/min

Chapter 1

INTRODUCTION

1.1 Introduction

Manufacturing Technology has been classified into three categories based on the fabrication technique involved. These processes are Subtractive, Formative and Additive. Subtractive manufacturing processes is one in which the material is subjected to be removed get the desired or final shape of the components with required parameters. Formative manufacturing process is one in which mechanical forces (like stresses and deformation) are applied to the material to bring the component to the required shape by deforming the material. Additive manufacturing processes is one in which material in the form of powders are added in a layer by layer fashion as well as treated at the same time (with heat source like laser, electron beam, etc.) to fabricate the component from a 3D model data.

In broad sense we can described Additive Manufacturing (AM) as the process of adding materials with the main objective of making objects from three-dimensional (3D) model data using the layer-by-layer principle. According to the ASTM Committee F12, **Additive manufacturing** is defined as a process of joining materials to make objects from 3D model data, usually layer upon layer as opposed to traditional processes based on subtractive manufacturing technologies.

In recent years, the use of additive manufacturing for a variety of purposes has expanded because of the advent of advanced additive manufacturing techniques which are being developed and getting approved. Today, additive manufacturing has been put to effective use to make a variety of products, these products includes:

- Medical Implants
- Orthopedic and Dental parts
- Electronic Devices
- Artistic Designs
- Jewellery Designs
- Commercial Lighting Products
- Videogame Avatars
- Engineered Foods etc.

Quality of products also depends upon the feedstock or raw material used. Powder quality control is one of the most essential factors to get parts with desired mechanical properties in Laser Additive manufacturing technique. Important advantage of this technique that it allows an efficient use of the raw material, as it has possibility to recycle and reuse unmelted powder. But the raw material properties may change due to successive reuse and might affect the mechanical behavior of parts. Metal powders are used to manufacture highly précised metal parts with a wide range of performance characteristics. Any metal can be made into a powder through processes including atomization, electrolysis, chemical reduction, and size .The production method can be chosen based on determined cost analysis, raw material supply, and the particle size and shape distributions most beneficial to performance. After that metal powder can be used for the fabrication of parts from simple single level shapes to complex multi-level shapes.

1.2 Additive Manufacturing Process

Additive Manufacturing is one of the fabrication technique which use layer-by-layer approach for joining of material or making it as 3D part and this 3D model is generated from CAD model and then it is being sliced in “n” number of layers and passed on to printer to print. Commonly Additive manufacturing is also known as 3D Printing and due to its layer wise process it is also have various other names like additive layer manufacturing, rapid prototyping and freeform fabrication. This technology got huge opportunity for research and development in almost all field starting from aerospace, medical, automotive to consumer goods.

1.2.1 Process Classification

The Standardized Definition given by F42 a technical committee on additive manufacturing in American Society for testing and Materials(ASTM) is the ‘process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to subtractive manufacturing methodologies’. Additive manufacturing committee also classified it in major Seven Process. Out of these seven processes only four processes are capable for manufacturing of metal parts.

1.2.2 Advantages and Disadvantages

Additive manufacturing is growing industrial sectors and reasons for the growth are its various advantages. Following are the few general advantages for the AM process:

Less Material Waste

Reduction in material waste great advantage for those industry which uses high cost material or super alloys and this reduction in wastage will help to reduce the cost without compromising with the mechanical properties or strength of the part fabricated. AM technology is very good example of bottom up approach instead of using large block to convert it into small components which increase the wastage.

More than 50% of material waste can be reduced through LMD process. It is very helpful to parts with high buy to fly ratio i.e. weight ratio between large block and finished part. AM efficiently help to reduce material use and cost.

Short Production Time

Another advantage to AM process is the reduced production time compared to the conventional manufacturing process, as AM is a single step process which helps to get finished part from the design given. On about 75% of time is reduced as compared to conventional production.

Customized Production

Additive Manufacturing is very much suitable for the small or medium size customization. Due to its layer-by-layer approach complex shaped parts that require difficult machining can also be printed in simpler way. In conventional manufacturing design change is difficult as it will give additional cost for the tooling but it is easier in AM to make design change as the only change to be made is in the input cad model.

Additive Manufacturing is a revolutionary technology so expectations and believes to the technology is much more but as of now AM have some disadvantages which also a big concern for the industry persons. Few generalized disadvantages are as follow:

Low Deposition Rate

AM have good production rate for low volume production as compare to conventional but as of now for high-volume production Conventional is far better than AM and many industries need complex geometry on high volume production rate.

Raw Material

Raw material available to the process is limited due to which many time industry have to compromise mechanical properties by choosing nearby material available. Limited material are available but still cost of those material is quite high, these prices for both machine and material may likely to reduce as the world accept this technology on large scale.

Quality & Reliability

AM is emerging technology and without proper research to the produce component industry cannot accept part for direct use to the high demanding industry sector. Proper research is needs to understand the process liability and structural properties.

1.3 Direct Metal Laser Sintering

1.3.1 Basic Principle of Sintering Process

The process of direct metal laser sintering creates surface layers or near-net-shaped by laser sintering of powder and substrate. There are three platforms i.e. building platform, dispenser platform and collector platform. Recoater is moving for spreading the powder from dispenser to building platform and for no collision of recoater to building and dispenser platform both are lowered by one-layer thickness. Dispenser platform rise to supply powder to recoater to take it to building platform when recoater is at right end and recoater move from right to left to carry powder. Excess powder carried by the recoater is then falls into the collector. Recoater homing position is left end. Scan head will move the laser two-dimensional across the cross section giving exposure to part area only by turning on and off the switch. Energy observed by the metal powder will generate the cure and sinter the already solidified areas below. Whole process is carried out to give the final product working layer-by-layer. Highly complex and accurate will be produced by the machine after few hours. Based on the working of the part post-processing treatment are carried out like tempering or surface treatment as per supplier material sheet or standard.

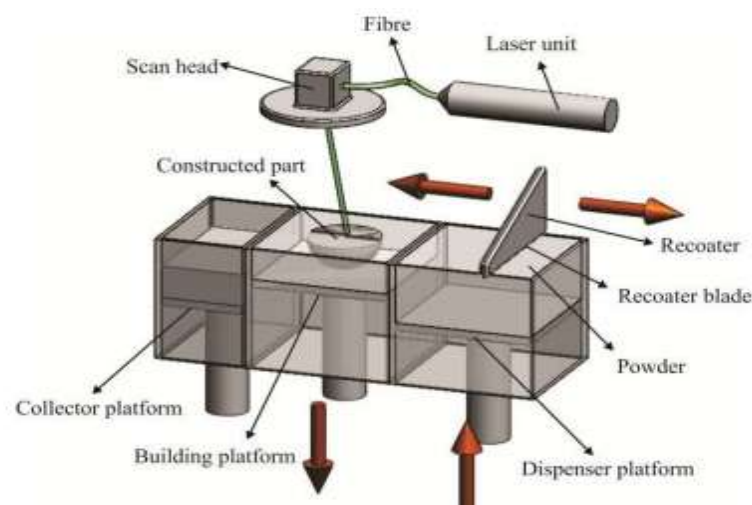


Figure 1. 1 Schematic diagram of the DMLS system

1.3.2 Laser Exposure Strategies

In the figure it has been shown the laser scanning of the top most layer out of the sliced layer on object. First layer will be exposed with Laser Powder (L_{pw}) and contour speed (C_{sp}). Diameter of the sintered zone is mostly higher than that of laser diameter or the curing zone (EOS,2009), so it is required to calibrate the dimensional error and laser beam must be shifted half the curing width from the contour to the inside, to make sure that sintering is done as per the original cad data. Correction or calibration of this position is called as Beam Offset. During Hatching, laser beam moves line after line to assure the sintering process as it maintains the temperature for long period. Distance between the lines is called as hatch spacing (H_s) and this $(1/4)^{th}$ of the laser beam diameter. As per the edge boundary given beam offset is again defined. Region will be not sintered or over-sintered if the value is higher or less than the correct value.

One of the other important parameter which can lead to distorted part or process interruption is the layer thickness. Too high layer thickness can lead to no optimum adhesion between the layer and this can lead to the crashed part as it will lead detachment of the layer below. If value is low then tearing of part may occur when recoater is moving.

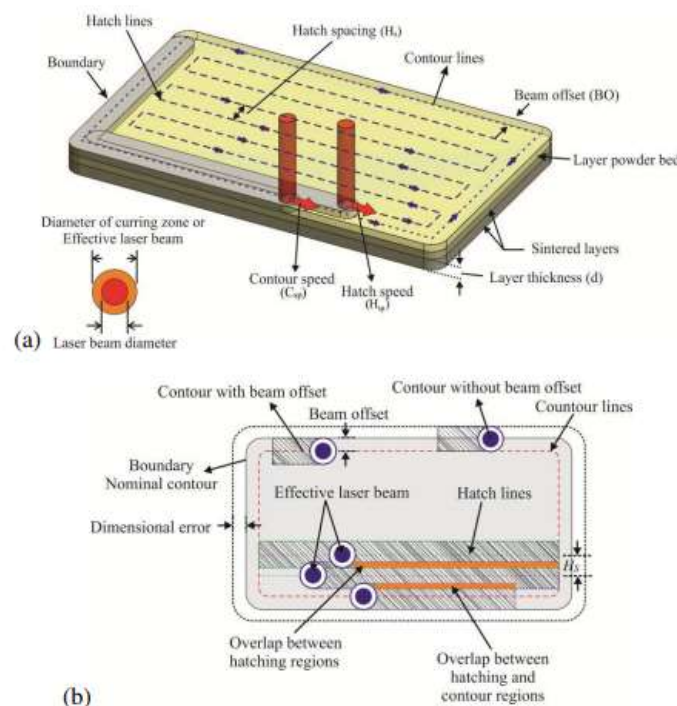


Figure 1. 2 Exposure strategies and process parameters

1.3.3 File Preparation and Fabrication

When a CAD designed object is to convert into the physical model it requires mainly two stages which includes data preparation and object fabrication by help of 3D printing machine. First of all part is designed by a modeling software. After that this model is converted to STL file format, and then the object is sliced into many layers where the parameters for components construction are also selected. At the final stage this final sliced file is given to the machine for printing. This whole process can be completed in few days or may be few hours. In this work, DMLS machine M290 and M400 was used for characterization of Ti64 and Inconel 718 powder respectively, both powders are from same source .i.e. EOS.

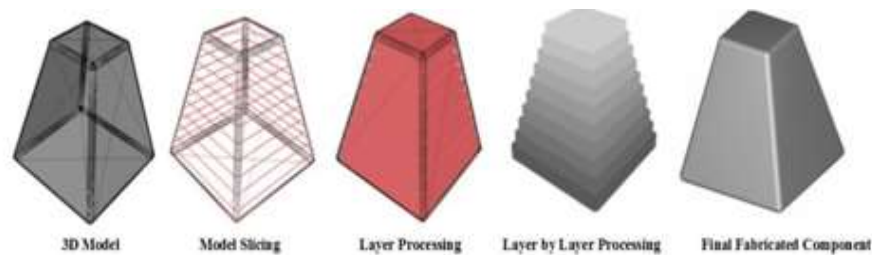


Figure 1. 3 File Preparation



Figure 1. 4 EOS M290

1.4 Powder used for this study

1.4.1 Nickel Based Super alloy

Now a day the biggest research field for additive manufacturing is aerospace and, in this sector, nickel based super alloys have great demand mainly for gas turbines. For aerospace, materials are selected for good mechanical properties at elevated temperature. In case of jet engines its operating temperature can reach up to 1300°C and these super alloys can match this requirement. Even these metals have high oxidation and corrosion resistance during long periods. Chromium (Cr) are added with the help of titanium to get the hot corrosion resistance. Because of the quality of these super alloys market price for these alloys are quite high, by AM wastage can be reduced to make it economical. Combustion chamber, exhaust nozzle of a jet engine are dependent on these super alloys. Nickel based super alloys consisted of two phases, a face-centered cubic (FCC)-nickel matrix and ordered distributed phase that is with FCC Crystal Structure (see Figure). Composition of 40-50 wt% Ni along with various alloying elements is generalized composition for these alloys.



Figure 1. 5 Schematic illustrations of the two main crystal structures in a nickel based super alloy: FCC γ -nickel (left) and (right)

Inconel 718- Inconel 718 is one of the most common and commercially known Nickel-based Super alloy used at lower service temperatures up to 750°C. It is also widely utilized nickel

alloy; about 50% to 70% times it is used in the jet industry. In 718 have a very high strength and corrosion resistance material exhibits high strength and corrosion resistance at temperatures up to 700 C. Its high strength resulting from both solid solution strengthening and precipitation hardening makes it a difficult material to machine.

Due to its low thermal conductivity and high hardness, tools wear out quickly when Inconel 718 is machined. However, these are some of the features that make the material so desirable in high temperature applications. Finding an alternative to subtractive manufacture high strength, high temperature materials are therefore an important topic of research in which AM is already becoming a practical option.

Table 1. 1 Chemical composition of IN 718

Elements	Min (wt. %)	Max (wt. %)
Nickel	50	55
Chromium	17	21
Niobium	4.75	5.5
Molybdenum	2.8	3.3
Titanium	0.65	1.15
Aluminum	0.2	0.8
Cobalt	-	1
Copper	-	0.3
Silicon	-	0.35
Manganese	-	0.35
Phosphorus	-	0.015
Sulphur	-	0.015
Boron	-	0.006
Iron	Bal	

1.4.2 Titanium

Density of titanium is half of the steel even though it has comparable strength to that of steel. It is used in chemical industries and even in some cases for jewelry also as it has very good corrosion resistance. Titanium is also used in medical industry for prosthetic elements as it has excellent biocompatibility. Pure Titanium at room temperature has hexagonal close-packed (hcp) crystal structure which after 883°C undergoes phase-transformation to body-

Centered Cubic (bcc) crystal structure. This temperature is known as Beta-Transus Temperature. Beta-Transus is defined as the lowest temperature in which the Titanium or Titanium alloy is 100% Beta.

The hcp phase often referred as Alpha-phase and bcc is referred as Beta-phase. Alloying elements in titanium can act as stabilizing agents either to alpha phase or either to beta phase. Alpha stabilizers on their addition will raise the beta transus temperature so that the alpha phase could remain stable and subsequently Beta stabilizers on their addition will decrease the beta-transus temperature. Some of the alpha stabilizers are Oxygen, Nitrogen, Aluminum and some of the beta stabilizers are Molybdenum, Vanadium. This Beta-Transus, Alpha and beta stabilizers form the basis for creation of Titanium alloys. Titanium alloys are generally classified into three types: Alpha alloys, Alpha-Beta alloys and Beta alloys. Alpha alloys contain neutral alloying elements (such as Sn) and/or alpha stabilizers (such as Al, O) only and their properties cannot much be influenced by heat treatment. Examples of Alpha alloys are Ti-5Al-2.5Sn, Ti-8Al-1Mo-1V, Ti-6Al-2Sn-4Zr-2Mo, etc. For alpha alloys, we can only do stress relieving and annealing. Alpha-Beta alloys generally contain a combination of alpha and beta stabilizers and are heat treatable to various degrees. Examples of alpha-beta alloys are Ti-6Al-4V, Ti-6Al-7Nb, Ti-6Al-6V-2Sn, Ti-3Al-2.5V, etc. Beta alloys are metastable and contain sufficient beta stabilizers (such as Mo, V) to completely retain the beta phase upon quenching, and can be solution treated and aged to achieve significant increases in strength. Examples of beta alloys are Ti-13V-11Cr-3Al, Ti-11.5Mo-6Zr-4.5Sn, Ti-10V-2Fe-3Al, etc.

Ti-6Al-4V- Ti-6Al-4V is the most common Titanium alloy, accounting for more than half of the total titanium usage. It is often called as the Work Horse of the titanium Industry. It is an alpha-beta alloy which can be heat treated to obtain the desired level of different mechanical properties. It is recommended for maximum use at a maximum service temperature of 350°C.

Grade 5 Ti-6Al-4V is known as the standard grade Ti-6Al-4V alloy. It is considered in those applications where a combination of high strength at low to moderate temperatures, light weight and excellent corrosion resistance are required.

Grade 23 Ti-6Al-4V is also known as Extra-Low Interstitial (ELI) or higher-purity version of Ti-6Al-4V alloy. It has lower specified limits on Iron and the interstitial elements Carbon and Oxygen. It has been the material of choice for many medical and dental applications due to its excellent biocompatibility. The ELI grade has superior fracture toughness, fatigue crack growth rate and better mechanical properties at cryogenic temperatures compared to standard

grade Ti-6Al-4V. The chemical compositions for Ti-6Al-4V Grade 5 and Ti-6Al-4V Grade 23 are given in tables 1 and 2 respectively.

Table 1. 2 Theoretical Chemical Composition of Ti-6Al-4V Grade 5

Element	Percentage
Carbon	0.25 max
Nitrogen	0.05 max
Oxygen	0.2 max
Aluminium	5.5 – 6.76
Vanadium	3.5 – 4.5
Iron	0.25
Titanium	Balance
Others	0.40

1.5 Powder Manufacturing Processes

Powder flow and spreading behavior is very much influenced by the process based on how the manufacturing of powder is done. This is influenced by the quality and size distribution of the powder. Out of all the metal powder producing technique gas atomization (GA), plasma atomization (PA) and plasma rotating electrode process (PREP) is the most common technique to produce high quality powders which will be suitable for powder bed fusion process. PSD (Particle size distribution) is controlled by the manufacture by using sieve analysis.

1.5.1 Gas Atomization

It is a common method for producing desired spherical particles. Raw material is introduced into the top furnace where the material is converted to the molten state. Molten metal then enters the chamber below the furnace through a broad open container one hole at the bottom, where it gets atomized with the help of high-pressure jet stream of inert gas. Solidified metal droplets are then collected at the bottom chamber. In aerospace industry the purity of powder is a big concern so, melting is usually performed by vacuum induction or electrode induction melting furnaces, Vacuum inert gas atomization (VIGA) produces refined and degassed melts where the melt pours through a broad open container one hole at the bottom to the atomization chamber. In case of Electrode induction melting gas atomization (EIGA) the raw material is used in terms of rod. In this case there is no use of any container or nozzle the melted droplets directly enter the atomization chamber. There is no contact with the melting crucible so the risk of contamination is minimized for the reactive alloy. This will result to spherical particles of wide range i.e. 0-500 μm . by controlling the gas flow the particle size distribution can be made narrower. Usually GA generates satellites which are surface irregularities consisting of smaller particles.

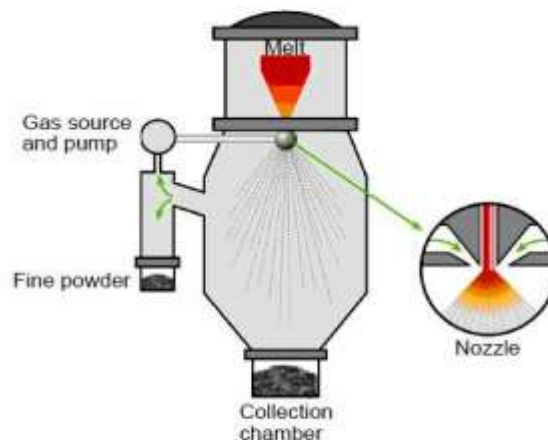


figure 1. 6 Basic illustration of the gas atomization process



Figure 1. 7 Illustration of a tilting crucible (left) and a bottom pouring crucible (right) used for a vacuum induction melting furnace Illustration of the induction coil used for an electrode induction melting

1.5.2 Plasma Atomization

Highly spherical particle of good quality can be produced by plasma atomization by using raw material with high melting point. Plasma torch is used to melt the feedstock which in terms of wire is fed into the chamber. Inert gas environment inside the chamber solidifies the metal droplets when it is freely falling into the chamber, there will be no impurities as atomized droplets never come in contact with any solid surface during solidification. PSD range for this process is between 0-250 μm .

Plasma Rotation Electrode Process- It is a powder production method similar to plasma atomization but there is mainly two differences i.e. PREP uses Bar as feed stock and material rotates as it is melted. Anyhow in this also the process is carried out in the inert gas environment. Due to the centrifugal force material atomizes and solidifies rapidly. Range for this method is between 0-100 μm .

Chapter 2

LITERATURE SURVEY

2.1 Powder Characteristics

While using the powder bed fusion technique for production of part it very necessary to understand and assess the knowledge about powder characteristics and its impact on the process. Morphology, Particle size distribution, flow ability, apparent density, bulk properties and porosity are the powder characteristics are important to ensure powder performance in the final structures. Spreading of powder is also important to have uniform layer thickness. ASTM International has provided a technical standard guide to characterize metal powder in purpose for additive manufacturing (ASTM International, 2014).

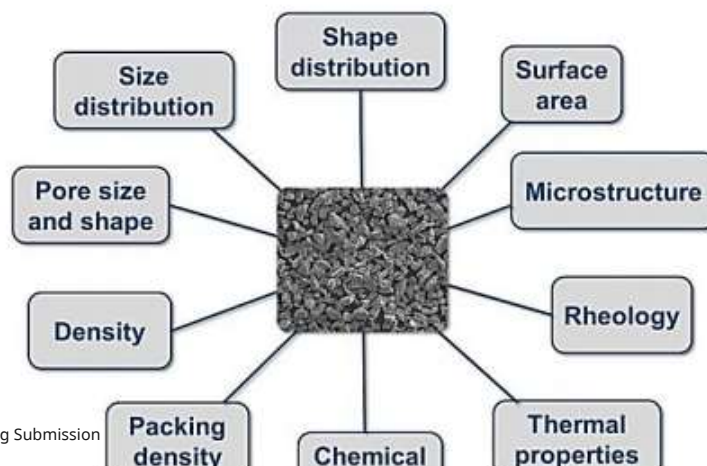


Figure 2. 1 Typical powder characterization parameters

Table 2. 1 Some Material properties and their significance to laser sintering process

Property	Significance to laser sintering process
Particle Size Distribution(mean and width)	Sinter ability, Packing efficiency, surface roughness
Particle Shape	Packing efficiency
Apparent Densities	Packing efficiency
Rheology	Uniform spreading of powder layer
Burnout	Decomposition temperature and ash content of binders
Specific heat, thermal conductivity	Heat transfer in powder Bed

Quality Assessment of Powder- few are the flowing powder characteristics that gives indications to the high-quality powder for DMLS.

- Large number of spherical particles
- Fine surfaces
- Very less number of satellites
- Low internal porosity
- Few surfaces pore
- Narrow size distribution
- High purity

2.1.1 Morphology

One of the important characteristics that influence the packaging density & flow ability of direct metal laser sintering is morphology of the powder. This characteristic is based on how the powder is being manufactured. PA and PREP manufactured parts shows high quality powder with less irregularities. Theses satellites are formed due to the difference in the solidification rate and are more formed in GA. Particle morphology can be studied through scanning electron microscopy

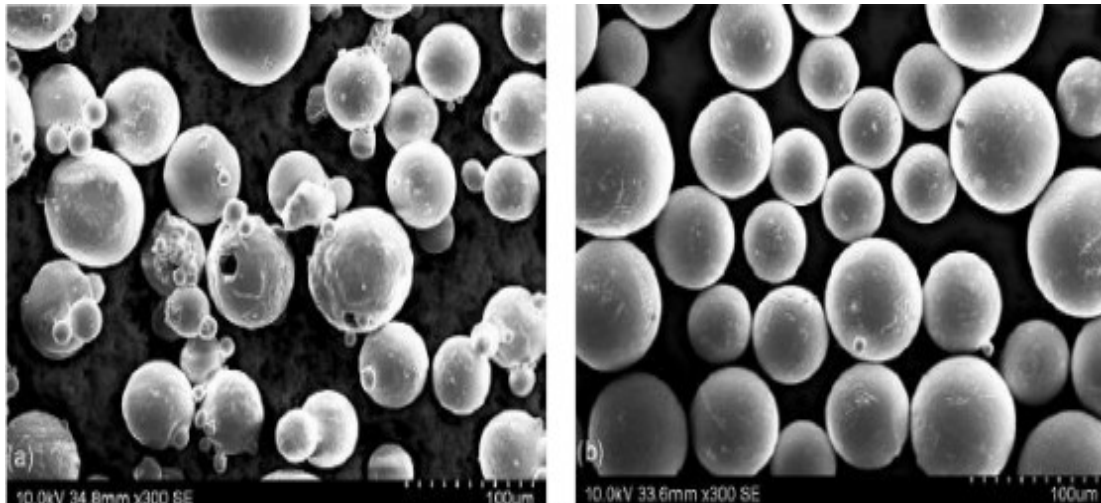


Figure 2. 2 SEM images showing comparison in particle morphology of GA (left) and PREP (right) produced Ti6A4V powders

2.1.2 Porosity

Porosity of powders both internal and on particle surface is disturbing and unwanted characteristics that will affect the surface properties of the part. This is also be influenced by the technique opted by the powder manufacturer. GA has high porosity as compare to the PREP manufacturing method. This degree of porosity can be explained by entrapped gas inside the particle due to the nature of the GA process. Surface pores present in GA are usually non-existing for the PREP powder. Using optical microscopy internal porosity can be viewed with polished cross-section.

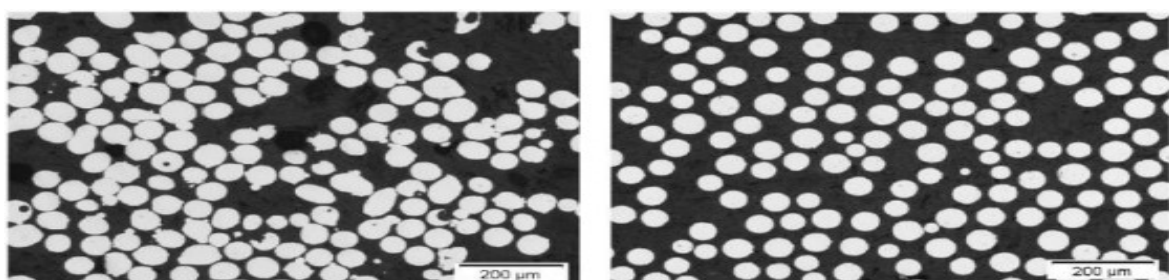


Figure 2. 3 OM image of polished cross-sections of GA (left) and PREP (right) produced Inconel 718 powders viewing internal porosity

2.1.3 Size & Size Distribution

There are two main types of particle size distributions considered depending on the technology in additive manufacturing: -

- Powders below 50 μ are used for most of the powder bed systems. In this case, fine powder particle which are below 10 or 20 microns should be avoided, as they are detrimental to the powder flow ability.
- Powder of sizes from 10 microns to 70 microns is used in DMLS and from 70 microns to 150 microns is being used in DED process.

The particle size distributions are a factor indicating what sizes of powder is present and in what proportions i.e. the relative particle amount as a percentage of volume where the total amount of particles is 100 %) in the sample particle group to be measured. The graph will show the percentage amount of sizes present in every interval. Alternatively, cumulative distribution expresses the percentage of the amounts of particles below a certain size.

A common approach to define the distribution width is to refer to three values on the x-axis (volume %):

- the D10 i.e. the size where 10 percent of the population lies below D10
- the D50, or median, i.e. the size where 50 percent of the population lies below D50
- the D90, i.e. the size where 90 percent of the population lies below D90

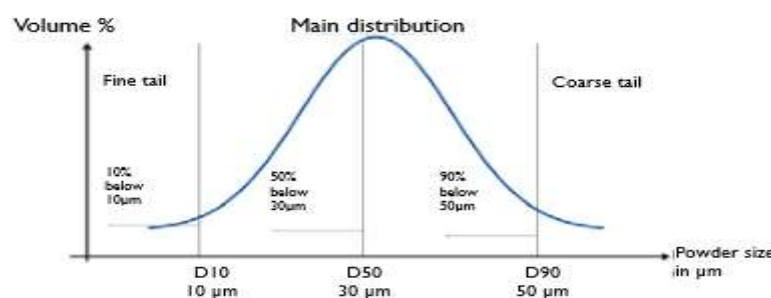


Figure 2. 4 Example of D10, D50 and D90 on a PSD curve for 10-70micorns powder

Depending on the method of production of powder the sizes of powder will vary & desired particle size range depends on the process of application, generally in case of powder bed fusion technique it ranges from 10-65 μ m.

Particle size affects the flow, spread and packing density (EPMA, 2015). PSD can also affect the efficiency & powder with a broad size distribution and using spherical fine powder can improve forming density. There will be decrease in efficiency if too large particles are available due to a resulting powder beam focus larger than laser beam focus, consequentially not able to melt all powder. Dynamic image analysis, laser diffraction and dry sieving are the few common techniques to study the particle size distribution of the powder. Many techniques operate by taking physical assumptions for their shapes and irregularities.

2.1.4 Rheology

Flow of powder particle is very important aspects for additive manufacturing because it affects the spreading and packing density of the powder over platform which ultimately affect the production rate for the industry. Flow of powder is due to the friction between the particles that's why particle size and distribution affect the flow ability. So, High Degree of presence of satellite will affect the flow due to the mechanical interlocking between the surface irregularities. Moisture is another reason for bad flow ability as antiparticle friction will be greater due to the high moisture content and may result in large particle agglomerates.

- Hall Flow meter- There is ASTM standard to measure the flow ability of powder based on the time taken by particular mass of powder to freely flow in to the container. The funnel in which this powder is flow through is called as Hall Flow meter funnel. ASTM Standard for this is ASTM B213. Mass of about 50g is placed into the funnel using a cone shaped tool and powder is released, time is calculated for the powder to fall into the container. We have considered Static flow of powder for the experiment. This method sensitivity is very low but it can be used to compare powder properties between each other.
- Carney Flow meter- This method covers the measure of a flow rate of those powders which are not easily flow able from the Hall funnel of Test Method. This is also a standard method for those powders which don't flow freely. ASTM for Carney flow is ASTM B964.

2.1.5 Bulk Properties

Apparent Density - It is a measured physical characteristic related to a powder's ability to pack. Apparent density is the property important to see the packing of powder when it is being spread over the platform using re-coater. It shows the relation of powder mass to freely flow over the empty or hollow space. This property is also measured by using hall flow meter funnel to freely flow powder through it and using ASTM B212 Standard. When the powder doesn't have ability to flow freely through hall flow meter, carney flow funnel is used for the apparent density calculation.

Tap Density- Another factor influencing the density is the tap density of the powder which is the ability of mass to fill the vessel after certain number of taps to the vessel. This will give increased density as compare to the powder's bulk density. This will include handling, transporting vibrations and between the two values will indicate how sensitive a powder is to possible vibrations.

2.1.6 Powder Impurities (Contamination)

Additive manufacturing is going to become one of the established production technologies. However, as it is still new the required process steps have not been properly defined yet. As no industry-wide standards describes or explain about the quality control process. One of the key parameter established till now is the particle size distribution of the powder used for AM. However, Particle size is not the only characteristic used for quality control. To have a check on the quality and purity of raw materials, suitable processes need to be suggested which can give the content of various "foreign" elements. And it should be properly monitored to ensure a high-quality end product.

Following are the various foreign Elements which can influence on the material properties and the most important is non-metallic elements and their effects which are described:-

Carbon [C]: The carbon content can affect various physical parameters. This ferrous alloy contains between 0.0002% and 2.06% of carbon. The higher the carbon content, the lower the melting point. Moreover, brittleness and hardness increase with the carbon content.

Sulfur [S]: If the alloy contains sulfur, this increases the machinability of the steel, i. e. the material's suitability for being treated by methods like drilling or milling. The higher the sulfur content, the lower the ductility.

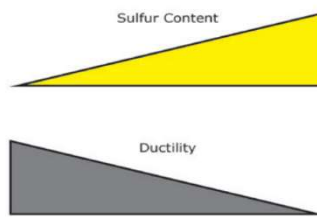


Figure 2. 5 Sulfur content v/s Ductility

Nitrogen [N]: The nitrogen content may be divided into desired and undesired content. There are some special applications which permit a high nitrogen concentration. In these cases, its chemical form has to be taken into account. Nitrogen in its elemental form is localized along the grain boundaries and influences the ductility of steel in a significant way. The nitrogen content which is bound to other elements is usually not considered important.

Oxygen [O]: Oxygen is a so-called steel parasite because it makes the steel brittle and causes ageing brittleness.

Hydrogen [H]: Hydrogen in steel causes the mechanical stability to degrade. Hydrogen embrittlement is widely feared because it may cause considerable technical and economic damages. It means that the protons attach themselves to the metal matrix which may lead to cracks in the steel.

Titanium

Hydrogen [H]: Has the same effect on titanium as on In718. Hydrogen may influence the formation of mixed phases in titanium alloys.

Nitrogen [N]: Nitrogen increases the brittleness of titanium.

Oxygen [O]: Even smallest amounts of oxygen have a considerable effect on the toughness or hardness of titanium.

Foreign elements such as Oxygen, Nitrogen, Carbon and Sulphur are measured by combustion and fusion techniques, as they may affect significantly material properties depending on alloys. With the gas atomization process, all powder particles have the same chemical composition but finer particles tend to have higher oxygen content due to the higher specific surface.

- The chemical composition will influence in particular:
- Melting temperature (Nano scale difference)
- Mechanical properties
- Weld ability

2.2 Product Quality

2.2.1 Part Density

Density of an additively manufactured refers to the porosity in the fabricated part. Some parts are required to be highly dense because of the application of that part. Parts should be able to operate under cyclic loading and must be dense or strong enough so as not to fatigue or fail under load. Most of the additive technologies can deliver are able to deliver part with densities of 98% and higher. In an additive build, usually during a laser sintering process, the layer-by-layer melting can be incomplete before subsequent layers are added on. This incomplete melting results in holes, cavities, or pores. Those areas of the part with the pores could more easily fail under load than other areas of the part. Depending on the part's application, such failure could be minor or catastrophic. Some of the factors contributing to porosity include powder particle size, build chamber temperature, and laser speed and temperature. Additive manufacturing vendors are working on ways to control porosity and density in those parts that require it.

2.2.2 Hardness

Resistance offered by the material to plastic deformation is known as Hardness. Hardness is an important factor when we talk about the service condition on which the part will be subjected to. It is necessary to study about hardness of the part produced.

- Quality Assessment of Powder- few are the flowing powder characteristics that gives indications to the high-quality powder for DMLS.
 - Large number of spherical particles
 - Fine surfaces
 - Very less number of satellites
 - Low internal porosity
 - Few surfaces pore
 - Narrow size distribution
 - High purity

2.3 Survey of Literature in the area of powder characterisation

Sames W J et al [1]: the techniques for producing metal parts are explored, which focus on the science of metal AM processing defects, heat transfer, solidification, solid-state precipitation, mechanical properties and post-processing metallurgy. Detail analysis of metal AM techniques are compared, with respect to their of the strengths and limitations. The author of this paper also mentions that quality of the feedstock that is used in the AM process is important for the quality final part with good mechanical properties. Quality of feedstock. i.e. metal powder used in Direct metal laser sintering is determined by size, shape, surface morphology, composition and amount of internal porosity. Physical variables are equally important such as flowability and apparent density. Atomization technique also somehow creates variations in the powder.

A. B. Spierings et al [2]: Author mention that particle size distribution not only affects the density but also the surface quality and the mechanical properties of the part. Study mention in this paper compares three different sizes of particle distribution depending upon layer thickness & scan speed. Two-layer thickness considered here are 30µm and 45µm. The surface roughness of the part can be improved by blasting. They have also concluded that the scan surface qualities, as well as the mechanical properties, are dependent on the particle size distribution. to optimize a powder material for SLM processes, the following aspects have to be taken into account.

- Relation of part density to the scan speed
- Scan surface quality, affecting the quality of the powder layers and the part itself
- Mechanical parameters

Fine particles are helpful to have a highly dense part as it can be easily melted and large particle size helps to maintain the specified layer thickness. Author also suggested that proper care or analysis should be done for better surface roughness and part properties.

Slotwinski, J. A et al [3]: Author has discussed the metrological challenges or issues related to the metal powder used in Additive manufacturing (AM) processes. Challenges associated with the measurement and characterization of the metal powders used for AM systems are also discussed. This paper describes the technical challenges and needs for characterizing metal AM powders. Every powder batch must be characterized according to the ASTM F3049 for good part quality. Author suggested various technical challenges which gives the justification for the need of powder characterization, including challenges associated with the measurement of AM metal powders with the relationship between powder properties and part properties.

Clayton, J. et al [4]: Authors have discussed powder properties are sensitive aspect for the additive manufacturing process which demands good consistency and repeatability. Not only particle size is a critical quality rheology of powder also plays equal role. Hall Flow test or Carney flow test describes the flowability behaviour of the powder. Authors have concluded that Powder flowability is the ability of powder to flow in a desired manner in a specific piece of equipment. Successful processing demands that the powder and the process are well matched. It is not uncommon for the same powder to perform well in one unit operation but poorly in another. This means that multiple characterization methods are required, the results

from which can be correlated with process ranking to produce a design space of parameters that correspond to acceptable process behaviour.

A. Strondl et al [5]: author have discussed that powder handling and their characterization are important issues for component made by using metal powder. By controlling the feedstock characteristics part or component properties also can be controlled. Parallel powder flow characteristics were also examine by the authors and later they concluded that particle shape have effect on flow of metal powder. They have compared the powder properties for SLM and EBM additive manufacturing processes. They have also concluded that there was not much difference in powder properties as compare to fresh powder and mechanical properties of part made by SLM have more influence due to the oxygen and porosity.

2.3.1 INFERENCE FROM THE LITERATURE

Additive Manufacturing is a latest technology and a lot of work is going on and a lot of work still needs to be done. The effect of powder on material properties is also one of the topics for research but there is not much work on the successive build cycles. Ti6Al4V and Inconel 718 are most extensively used metal powder and these are not much work on these powder. There is not much work done how to distinguish and evaluate powder characteristics for the powder which effect directly or indirectly on part properties. There is not much evaluation done to correlate the powder characteristics to part density, hardness and microstructure. The aim is to identify and discuss any and correlation between the characteristics of powder and final part quality.

2.3.2 PROBLEM STATEMENT

The main objective of this thesis work is the Powder Characterization and effect on powder after multiple uses in Laser Additive Manufacturing.

The objectives of thesis work include:

1. To understand the relationship between material properties and powder properties.
2. Effect on powder after multiple uses in SLM setup.

3. Reusability of powder
4. Analyzing Apparent Density and Part density after every build cycle
5. To study the hardness of the part
6. Health, Safety and Environmental issues
7. Effect of Oxygen and Nitrogen on properties of titanium alloys & Inconel 718

Chapter 3

Experimental procedure & Methodology

3.1 Materials & Methods

In this Chapter powder characterization technique is explained as per the ASTM Standard F3049. The methodology for Direct Metal Laser Sintering is explained followed by the quality assessment methods for the part fabricated.

3.1.1 Powder & Sheet Material

The materials used in this project are Inconel 718 alloy and Ti64 powder which are being used for additive manufacturing. These powders are dry and with no flow additives. The standard composition for the fresh powder is given in the following table. Studies have been made on two different powders with same commercial powder suppliers.

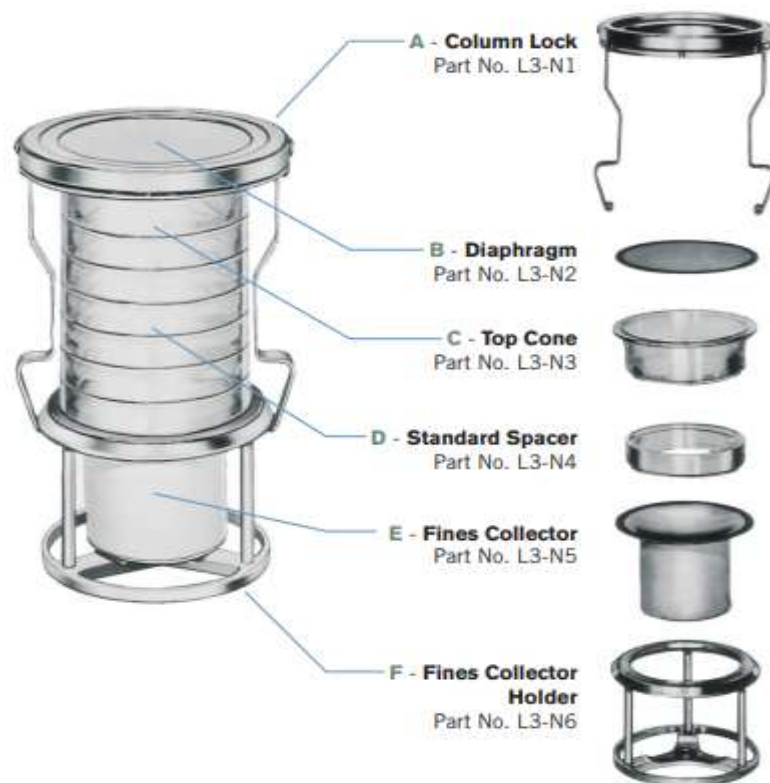
There was total six number of build for Inconel powder in EOS M-400 and four number of build for Ti64 in EOS M-290. Both powder characteristics were analysed after every build starting from virgin powder analysis. The current in-house material specification at wipro3d of Inconel 718 and Ti64 for direct metal laser sintering specifies required composition, manufacturing method, and hall flow rate and particle size distribution.

Test specimens were built with square cross section to study the microstructures in transverse and longitudinal directions. The square test specimens were having cross section of 4mm*4mm and length of 45mm. STL files shown in figure 4 were prepared for printing in Materialise MAGICS 22.01 RP Software. Parts were properly fixed and any bad edges or overlapping triangles were eliminated in MAGICS. Slicing of the STL files were done on EOS RP Tools. The SLI files were then imported to EOS Print through which the EOS M290 & EOS M400 DMLS Machine reads. The skin-core strategy was applied with ChessRotLX as the exposure type for melting of the powder in the machine. Each layer was scanned once. The laser scanned initially the edge area of each cross section and then the inside area with parallel alternating scan vectors at an angle of 67° to the previous deposited layer. The laser then scanned the boarder of each cross section in order to improve the surface roughness of the part and reduce the number of defects near the external surface of the component. The powder layer in the build plane is divided into several small squares and scanning in a straight line fashion within each square, but with a different orientation in the adjoining square. Each layer thickness was of 40 microns. Ytterbium fiber Laser with a power of 400W was used for DMLS process with the laser beam spot diameter of 80 microns having wavelength of 1070 nm. The default scanning speed kept was 1250mm/s. The build platform was kept at 35°C and the building chamber was first evacuated and then filled in with Argon gas. Oxygen level was kept below 0.13% for the building process. This was done to lower the risk of oxidation and fire inside the building chamber. After the parts were prepared, they were separated from the build platform by precisely cutting them using Electronica Wire EDM Machine.

3.2. Powder Characterization-

3.2.1 Particle Size Distribution

The PSD of the powders was measured by Sonic Sifter Separator. In dry powder sample will be poured from the top and putting coarse sieve at the top and fine at the bottom. There will be vibration given to the system by which powder will be distributed to different sieve and by weighing them we can calculate the PSD of powder. Each powder was tested three times. Horizontal pulse accessory is attached to get accuracy below 45μ .



Item	Description	Part Number	Unit Price
A	Column Lock	L3-N1	
B	Diaphragm	L3-N2	
C	Top Cone	L3-N3	
D	Standard Spacer	L3-N4	
E	Fines Collector	L3-N5	
F	Fines Collector Holder	L3-N6	
	Complete Stack Assembly*	L3-N7	

Figure 3. 1Sonic Sifter Apparatus

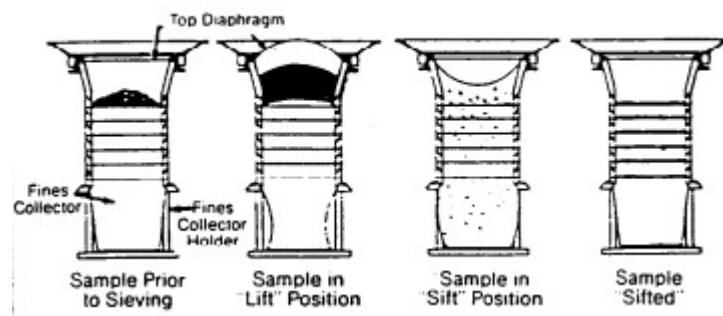


Figure 3. 2 Diagram of four conditions during sieving operations

3.2.2 Rheology

Hall Flowmeter

The Hall flow rate was measured by a static flow method. In the following figure the schematic illustration of the test setup is given. In this bottom of the orifice was blocked by finger and powder was poured into the centre of the funnel. Finger was removed from the orifice and simultaneously stopwatch was started manually. The mass of powder introduced to the funnel was of 50gm. five measurements of each powder and recorded and averaged out according to the ASTM standards (ASTM International, 2013a). All the measurements were performed by one operator. Protective gloves were used to avoid moisture content from finger to powder while blocking the orifice.

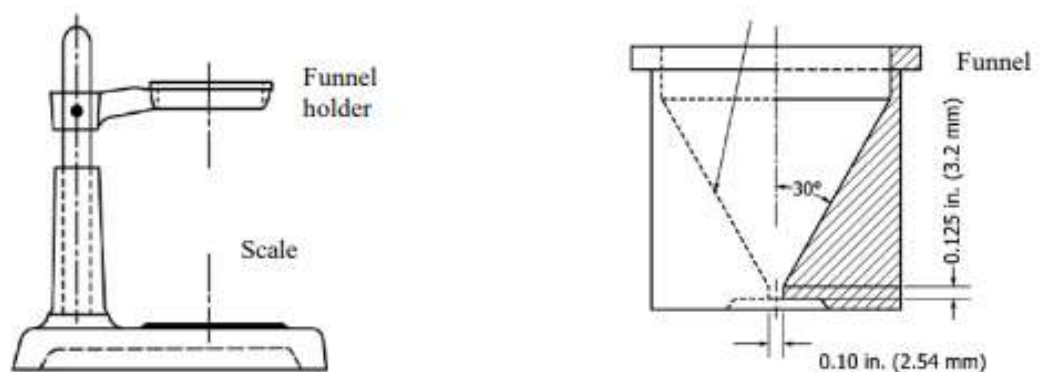


Figure 3. 3 Schematic illustration of Hall Flowmeter setup according to ASTM standard. Setup stand and bottom scale (left) and funnel (right) are the setup basis of the method

Carney Flowmeter

Similar process is carried in case but this test is for those metal powders that do not freely flow through the hall flowmeter funnel. Schematic diagram for this also is given below.

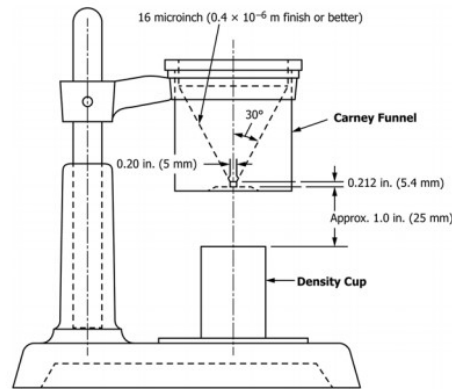


Figure 3. 4 Schematic illustration of Hall Flowmeter setup according to ASTM standard

3.2.3 Bulk Properties

Apparent Density is measured using a density cup of volume $(25.00 \pm 0.01) \text{ cm}^3$ which is measured using hall flowmeter setup for those powders which flow freely otherwise instead of hall flow, carney flowmeter is used. Empty Density cup is weighed before placing below the funnel. Powder is flowed freely into the density cup and allowed to overflow. Extra amount of powder on the density cup is levelled off using a nonmagnetic blade. Density cup is again weighed this time with powder and difference between the weights is recorded as powder mass. Powder mass is divided by cup volume and recorded as g/cm^3 .

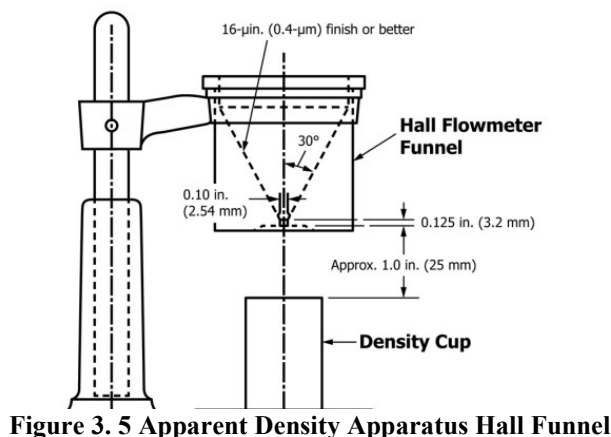


Figure 3. 5 Apparent Density Apparatus Hall Funnel

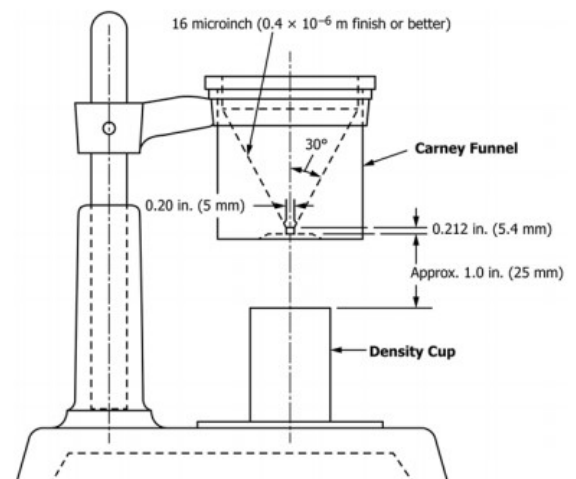


Figure 3. 6 Apparent Density Apparatus Carney Funnel

3.2.4 Powder Contamination

There are various ways for measuring the element concentrations and impurities and most of these require destruction of the sample. This is done to ensure that all relevant components of the analysed sample are released. In the present work, inert gas fusion technique is used for measuring and it also one of the destructive tests for the sample. This inert gas fusion technique offers a number of advantages. Average particle size for additive manufacturing technologies lies in between 5 μm and 150 μm . if the distribution sizes for the particles are in right proportion, then elemental concentrations can be found by combustion analysis.

The measurement of H/N/O/ cannot be carried out in two analysis. There are two modes for analysis in which one mode is Oxygen & nitrogen, and other mode is for hydrogen.

O/N/H Analysis

The sample is dropped into a graphite crucible. It melts due to the high temperature. Consequently, oxygen, nitrogen and hydrogen are released. Oxygen converts to CO on the surface of the hot crucible. The inert carrier gas removes the gases from the crucible.

A copper oxide catalyst converts CO to CO₂ which is detected in the infrared cells. An infrared ray with a specific wave length is used to excite the carbon dioxide molecules. The loss of energy, which is transformed to kinetic energy, is used to determine the exact oxygen concentration of the sample. The nitrogen and hydrogen content are measured in a thermal conductivity cell.

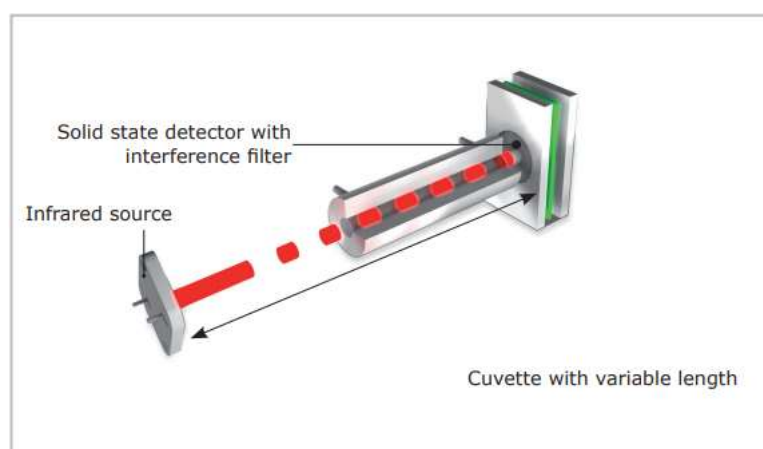


Figure 3. 7 Infrared Cells

The ELEMENTRAC thermal conductivity cell is based on a micromechanical silicon chip which is coupled to a membrane and works independently of a reference gas flow. If the thermal conductivity of the gas changes, for example, through nitrogen released from the sample, the heating capacity required for heating the membrane changes as well. This is indicated by a measuring signal. The method is robust and sensitive and it guarantees stable measuring results over a wide concentration range.

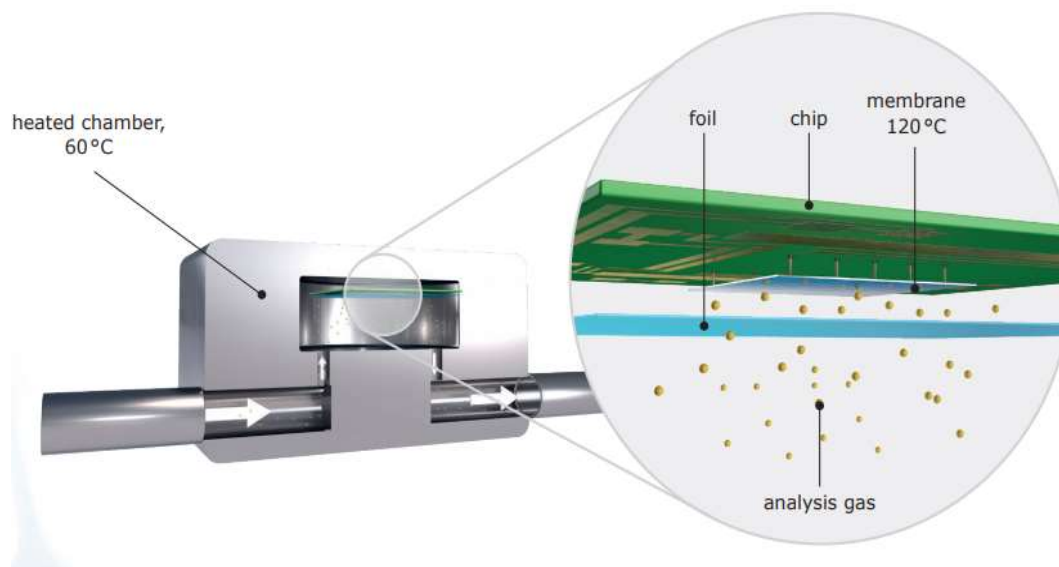


Figure 3. 8 Thermal Conductivity Cell

Table 3. 1 Different thermal conductivities

	M [g/mol]	Density [kg/m ³]	Coefficient of thermal conductivity [W/kW]
Hydrogen H ₂	2.02	0.08987	1.869
Helium He	4.00	0.17839	1.567
Nitrogen N ₂	28.01	1.2505	0.260
Argon Ar	39.94	1.7839	0.179

Following are 5 steps involved in the analysis process: -

Out gassing- The graphite crucible is heated without sample in the furnace to remove impurities present in it. The carrier gas supply to the furnace is in the purge mode. The time for outgassing and the furnace power are adjustable.

Purging- The furnace purging is continued after outgassing phase. The furnace is switched off.

Stabilizing- The carrier gas supply to the furnace is switched to the analysis mode. This phase is necessary to let the base line of the IR-cell and TC-cell stabilizes.

Sample drop- The sample drop mechanism rotates and the sample drops into the crucible.

Analysis- The furnace is switched on; the sample is melted and the signals from the detectors are processed. The furnace is switched off after the pre-set time expires, but the signals processing continues until the comparator level of all channels or the maximum analysis time is reached. In order to avoid unnecessary signals integration from the moment when furnace is switched on until the moment when the analytical gas reaches the detectors, the integration delay time can be adjusted.

3.3 Part Quality

3.3.1 True density

Archimedes method is used for calculation of true density of the part using **Contech** High Precision Balances with a specific density measurement device for solid materials. Figure 1 shows a schematic of the measuring principle indicating the measuring of the mass in normal air and in the fluid.

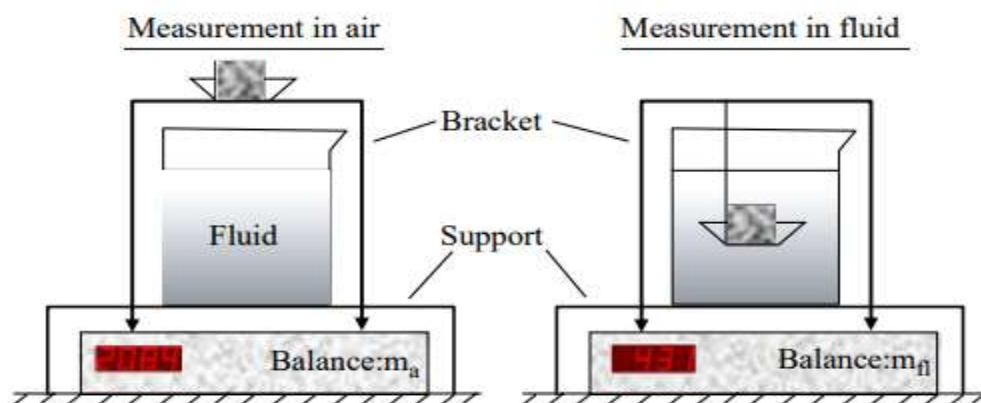


Figure 3. 9 Archimedes methods, showing the measurement of the mass in air and in the fluid

It has a measuring accuracy of ± 0.1 mg. The calculation of the density ρ_p of the part under consideration follows equation (1):

$$\rho_p = \frac{m_a}{m_a - m_{fl}} \cdot \rho_{fl} \quad \text{Equation 3.1}$$

ρ_{fl} is the density of the fluid, which is temperature dependent, m_a is the mass of the part in air and m_{fl} is the mass of the part in the fluid. A more correct calculation of the density takes into account the buoyancy of the part in air. This leads to a correction of equation (1), where ρ_{air} is the density of air (temperature dependent):

$$\rho_p = (\rho_{fl} - \rho_{air}) \cdot \frac{m_a}{m_a - m_{fl}} + \rho_{air} \quad \text{Equation 3.2}$$

Each sample was measured independently four times.

3.3.2 Hardness

The Rockwell hardness test is one of the indentation hardness tests used to measure the hardness for the given sample. Indentation hardness tests are measured for the deformation which occurs when the material is penetrated with a specific type of indenter. There are two levels of force which are applied to the indenter at specific rates and with specific dwell times. Rockwell hardness for the material is based on the difference in the depth of the indenter at two specific times during the testing cycle. Hardness value is calculated using a formula that was derived to yield a number falling within an arbitrarily defined range of numbers known as a Rockwell hardness scale.

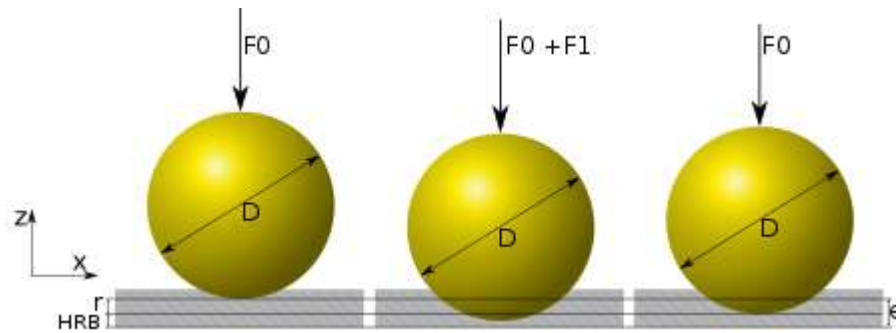


Figure 3. 10 Force Diagram for Rockwell Test

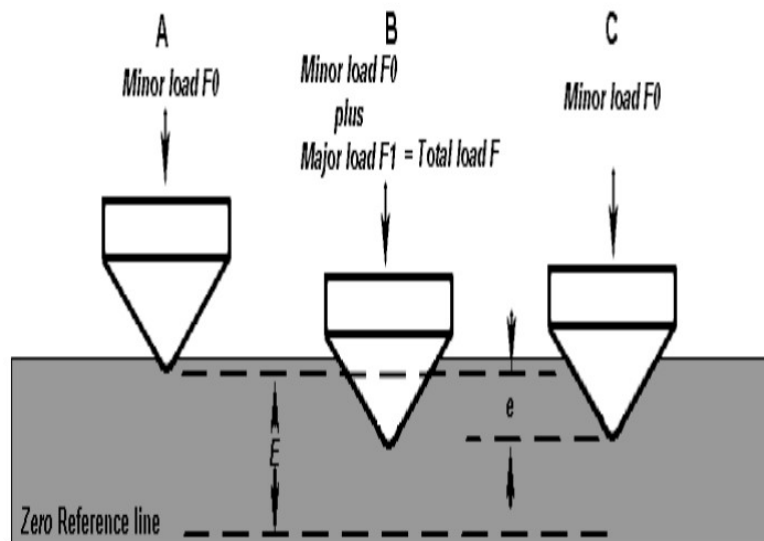


Figure 3. 11 Indentation

Hardened steel and hard alloys are tested in the Rockwell C scale and for the test block, Rockwell C is used, which is the hardness around HR61-68. After that, diamond penetrator is selected and the major load taken is 150 kg.

Chapter 4

RESULTS AND DISCUSSIONS

4.1 Particle size distribution

The results are given as a volume distribution of the particle sizes versus the Sieve sizes. The PSD of the In718 powder after reuses was analyzed and the results are shown in Figure 4. 1. The PSD of the powder after the second use is very similar to that of the virgin powder. Changes in the PSD were observed after 3rd build in case of In718 shown in Figure 4. 2

Table 4. 1 Particle size distribution In718

Sieve Size	Fresh	After FT	1st Part Build	2nd Part Build	3rd Part Build	4th Part Build	5th Part Build
0-20	22.51	21.33	13.91	4.84	8.45	7.56	2.93
20-32	44.32	44.36	45.35	40.12	39.16	38.15	37.63
32-45	22.08	21.56	19.76	36.14	33.18	38.54	40.74
45-53	8.24	6.67	10.32	10.73	15.72	12.81	17.19
53-63	0.38	1.23	1.13	4.71	1.76	1.44	0.82
63 & Above	3.65	5.36	12.9	3.47	0.73	1.49	0.59

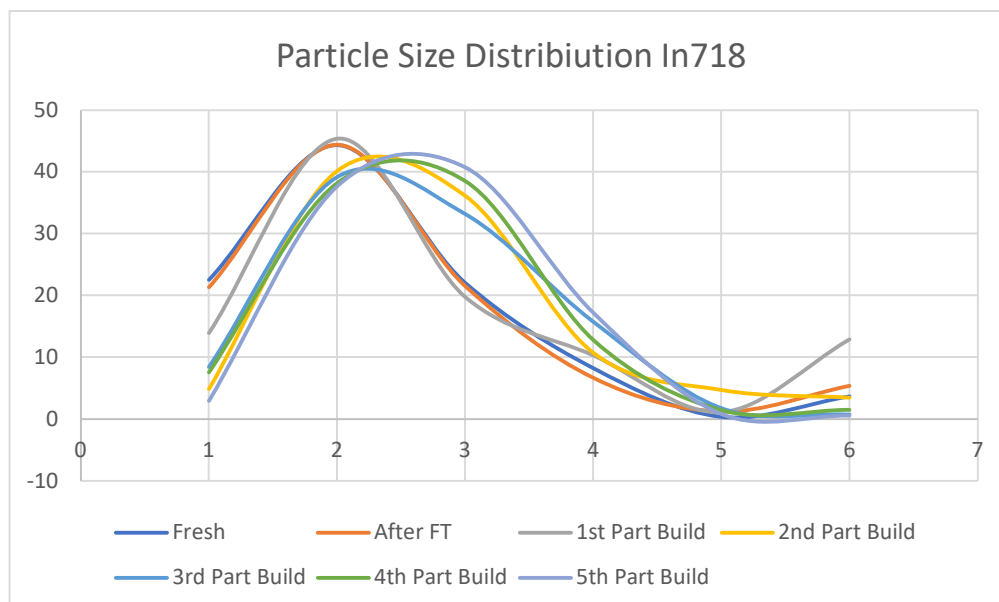


Figure 4. 2 Particle Size Distribution In718

There was very slight deviation in case of Ti64 as well.

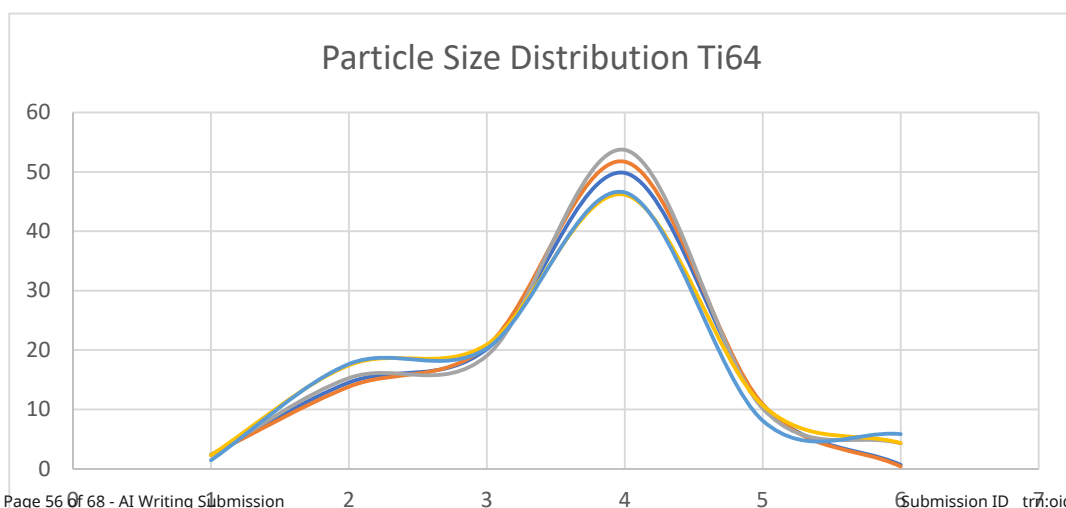


Figure 4. 3 Particle Size Distribution Ti64

Table 4. 2 Particle Size Distribution Ti64

Sieve Size	Fresh	After FT	1st Part Build	2nd Part Build	3rd Part Build
0-20	2.38	2.36	2.37	2.33	1.48
20-32	14.54	13.85	15.32	17.48	17.66
32-45	20.18	20.71	19.07	20.96	20.32
45-53	49.82	51.75	53.72	46.16	46.53
53-63	10.83	10.91	10.19	10.72	8.13
63 & Above	0.68	0.41	4.33	4.36	5.89

At the time of powder collection unmelted powder is blown off from the additively manufactured parts, some satellite particles are also attached and are taken away by the compressed gas flow. Because of which these satellite particles becomes less as powder is used again & again. Therefore, less satellite will be seen after use of powders. Agglomerated particles are also sieved out due to use of 80µm sieve after part building process, which result, the PSD becomes narrower with increasing powder reuse times. Particle size distribution also depends upon the fact that from where powder is taken.

4.2 Rheology

Flowability is an important parameter for powders as it simulates what happens during the powder-raking process. Among the various shapes of powder particles (irregular, angular, and spherical), only spherical-shaped particles can provide the highest flowability, as the particles of such powders have minimal contact areas with neighbouring powder particles, thus creating less internal friction force and requiring the minimal raking force in order to flow. The flowability of both the reused powder, were recorded in seconds per 50 g of reused powder for hall Flowmeter and seconds per 150g for carney flowmeter. As expected, powder flowability of both the powders was not good after 3rd build. Even after 3rd build, powder hall flow test was not able to be carried out as the powder required more than one tap.

After the reuse of powder, the particles start to deform slightly and even some powder stuck to others leaving less amount of spherical particle. The changes are beyond measurement errors and reproducible.

Table 4. 3 Flow ability for In718

S No.	Reuse Times	Hall Flowability (s/50 g)^a	Carney Flowability (s/150 g)^a
1	Fresh	14.32	0.06
2	After FT	13.23	0.06
3	After 1st Part	N/A	0.06
4	After 2nd Part	N/A	0.06
5	After 3rd Part	N/A	0.07
6	After 4th Part	N/A	0.07
7	After 5th Part	N/A	0.07

In addition, used powder may come into contact with air and absorb some amount of moisture during the powder-recovery process like sieving, reloading and storage. Moisture strongly effect the flow characteristics as it can be seen after 1st part was made.

Table 4. 4 Flow ability for Ti64

S No.	Reuse Times	Hall Flowability (s/50 g)^a	Carney Flowability (s/150 g)^a
1	Fresh	14.32	0.00
2	After FT	13.23	0.09
3	After 1st Part	N/A	0.12
4	After 2nd Part	N/A	0.13
5	After 3rd Part	N/A	0.07

4.3 Apparent Density

The measured apparent density for Both In718 and Ti64 powders build wise is shown in Table. The results show that there is an increase in apparent density for In718 after 3rd build and in case of Ti64 there was not much change in the apparent density. The powder with the lowest ability to freely pack will have lower A_D . For Ti64, the ability to freely pack a volume will be the same or almost equivalent, while in case of In718 it is better after 3rd build.

Table 4. 5 Apparent density In718

S No.	Reuse Times	Powder Weight	Apparent Density
1	Fresh	68.604	2.74416
2	After FT	68.5971	2.743884
3	After 1st Part	68.6012	2.744048
4	After 2nd Part	68.5983	2.743932
5	After 3rd Part	87.5616	3.502464
6	After 4th Part	87.4851	3.499404
7	After 5th Part	87.7821	3.511284

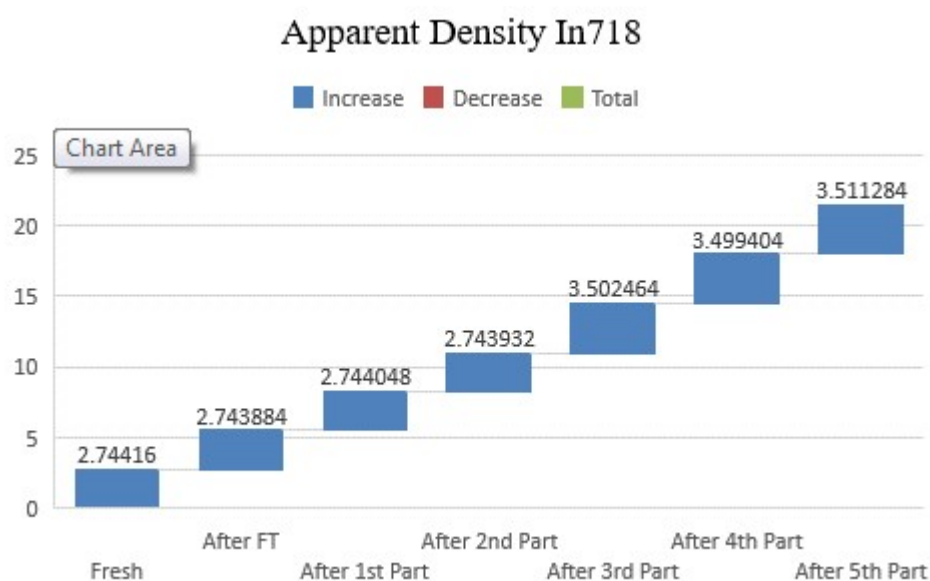


Figure 4. 4 Apparent Density In718

Table 4. 6 Apparent Density for Ti64

S No.	Reuse Times	Powder Weight	Apparent Density
1	Fresh	63.0079	2.520316
2	After FT	63.0244	2.520976
3	After 1st Part	63.7927	2.551708
4	After 2nd Part	59.3802	2.375208
5	After 3rd Part	61.9396	2.477584

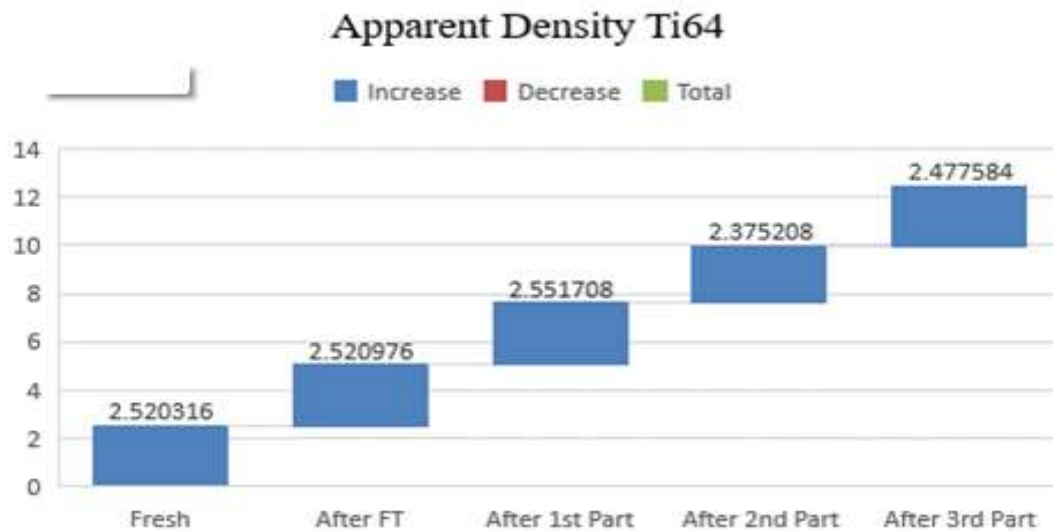


Figure 4. 5Apparent Density Ti64

The measured conditioned bulk density (CBD) from the LPW Apparent Density equipment, all show a higher value if compared to the apparent density which was expected. A lower density difference gives a powder which is less sensitive to vibrations. The best results are shown by Ti64 which has very less deviation. In718 powders show very similar results with a slightly larger difference in density.

4.4 True Density

Part Density for additively manufactured part for both In718 & Ti64 was measured using the Archimedes method. Not much comparative differences were found for densities after a number of builds for both IN718 and Ti64. Highly dense parts depend on the powder characteristics and the layer thickness. EOS Standard process parameter of 90micron for IN718 and 40micron for Ti64 were used for part fabrication.

Table 4. 7 Part Density for In718

S No.	Mass in Air	Mass in Water	Density
B1	10.4956	9.2126	8.1562
B2	10.3786	9.11	8.1632
B3	9.2238	8.0973	8.1637
B4	10.2898	9.0328	8.1623
B5	9.9165	8.7031	8.1405

Table 4. 8 Part Density for Ti64

Sl No.	Mass in Air	Mass in Water	Density
B1	5.4258	4.4833	4.2886
B2	5.3959	4.4651	4.2789
B3	5.8037	4.4742	4.3702

4.5 ONH

Figure 4.6 shows the nitrogen and oxygen content results for In718 after five numbers of builds. The oxygen levels for the powder are increasing on ppm scale and it increased from 141.8ppm to 199ppm. Oxygen level for In718 is well within the boundary. Results from the first study showed that powder was within the limits for oxygen over 5 builds. Nitrogen levels also showed a gradual increase after 4th build but staying well within the 500-ppm maximum specification and coming above the 150-ppm maximum for initial few builds. Under normal build conditions, the dispenser is refreshed with virgin or unused powder once the level in the dispenser begins to deplete but here, for top-up, last batch of powders was used.

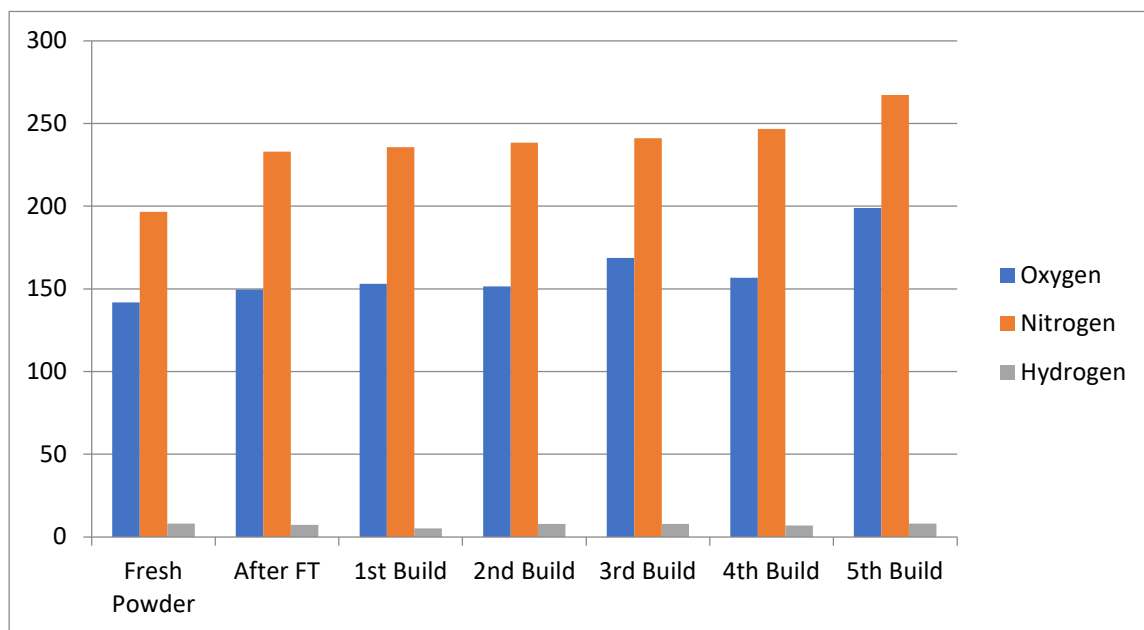


Figure 4. 6 ONH Content for IN718

Table 4. 9 ONH Content for In718

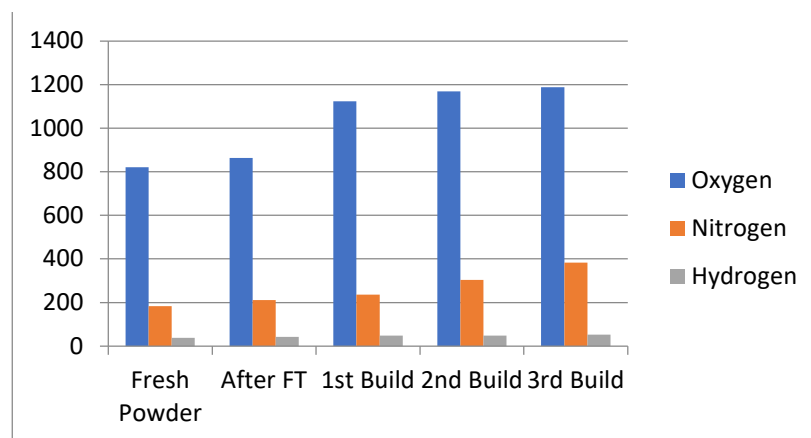
Contamination	Fresh Powder	After FT	1st Build	2nd Build	3rd Build	4th Build	5th Build
Oxygen	141.8	149.5	153.1	151.5	168.8	156.8	199
Nitrogen	196.5	233	235.7	238.4	241.2	246.7	267.2
Hydrogen	8.1	7.3	5.2	7.8	7.8	6.9	8.1

The increase of oxygen content in the reused powder happens due to the pick-up of oxygen from air when powder is exposed to air; this exposure to air is also due to the unmelted powder blown-off during the sieving process of additively manufactured parts. Pick-up of oxygen totally depends on the exposure time and the humidity of the environment.

Table 4. 10 ONH Content for Ti64

Contamination	Fresh Powder	After FT	1st Build	2nd Build	3rd Build
Oxygen	820.2	863.7	1123.8	1168.8	1188.3
Nitrogen	183	211.5	235.7	303.4	383.2
Hydrogen	38.1	42.3	48	47.7	52.1

Figure 4. 7 ONH Content for Ti64



4.6 Hardness

Hardness result for both the material is shown in following table and there was not much change found in the values for hardness. All specimens were heat treated as per the given EOS data sheet.

Table 4. 11 Hardness In718

S No.	1st Indentation	2nd Indentation	Hardness
B1	44	42	43
B2	43	42	42.5
B3	44	43	43.5
B4	43	44	43.5
B5	42	43	42.5

Table 4. 12 Hardness Ti64

S No.	1st Indentation	2nd Indentation	Hardness
B1	34	35	34.5
B2	36	35	35.5
B5	35	36	35.5

Chapter 5

CONCLUSIONS

- PSD becomes narrower with the increasing reuse time of powders and a few unburnt particles and satellites formed will be removed by sieve. As satellites were found, it can be assumed that particles are no longer smooth and it was visible before sieving of powder was done.
- Hall flow behaviour for both the powders i.e. Inconel 718 & Ti64 after 1st build not good as more than one tap was required and as per standard carney flow test was done.
- Flow behaviour in case of carney was constant as the number of builds increases; this is most likely due to narrowing of the PSD by gradual removal of smaller particles. Powder morphology does not significantly change over the entirety of the study. Occasional misshapen and agglomerated particles occur within the bulk but at a very low occurrence, and the effect does not seem to be cumulative.
- Ability to freely pack in a defined volume is increased after 3rd build in case of In718 but for Ti64 there was not much difference found.

- In the studies, there was a general increase in oxygen and nitrogen levels observed but in case of hydrogen it has very less effect. These impurities most likely appeared when particles are close to the melt pool areas which are partially melted. The level of impurities for both the powder was within the limit as given in the powder mill report. ONH content in the powder will be within acceptable boundaries if virgin powder is topped into the dispenser.
- The changes observed over the period of the studies do not seem to affect much the powder properties after 5 and 3 builds of IN718 and Ti6Al4V respectively, therefore the material parameters are still valid for the powder from start to end.
- Inconel 718 powders can sustain a large number of cycles without change in powder and subsequent build chemistries.

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